Predicting Housing Affordability Stress Across Greater Boston ZIP Codes

Tanmay Demble Samuel Seelan Ekrem Karakas

Northeastern University DS 4420 Machine Learning

Motivation

Housing affordability in Greater Boston has deteriorated sharply since 2020. Rising prices, constrained supply, and volatile mortgage rates have pushed homeownership out of reach for growing numbers of residents, but the pressure is not uniform across neighborhoods.

Goal: Predict which ZIP codes will experience affordability stress *before* it materializes, giving policymakers and prospective buyers an early warning signal.

Definition: A ZIP – month is *high stress* when the income required to buy a median home falls in the **top 25%** of all observations in the dataset.

Data

- **Zillow ZHVI:** Monthly ZIP-level home values across the Boston metro area, seasonally adjusted
- **FRED:** 30-year fixed mortgage rate, national unemployment rate, and Consumer Price Index
- **Zillow market signals:** Metro-level inventory counts and median days to pending sale
- **Target:** Zillow income needed to purchase, thresholded at the 75th percentile for binary labels

Dataset: Link to dataset

Three Modeling Approaches

1 SARIMAX

Time Series Forecasting

Order $(1,1,1)(1,1,1)_{12}$ selected via ACF/PACF across six representative ZIP codes. 30-year mortgage rate as exogenous regressor. Trained through 2022; evaluated on 2023 – 2024.

2 MLP Classifier

Binary Stress Prediction

Built from scratch in NumPy. 284 inputs → 32 (ReLU) → 16 (ReLU) → 1 (sigmoid). Learning rate 0.005, 600 epochs, binary cross-entropy loss. 3:1 class imbalance corrected via downsampling.

3 Bayesian Regression

Interpretable Inference

Implemented in R with `brms`/Stan. Predicts income needed to purchase a home with full posterior uncertainty over each predictor. Trained 2018 – 2023; evaluated 2024.

References

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[3] S. Muralidharan et al. (2018). Analysis and prediction of real estate prices: A case of the Boston housing market. *Issues in Information Systems*, vol. 19, no. 2, pp. 109–118.

[4] B. Zhang (2023). Analysis and forecast of influencing factors of house price in Boston. *Proc. DAML 2023*, pp. 107–113.

[5] R. Gupta, A. Kabundi & S. M. Miller (2011). Forecasting the US real house price index. *Economic Modelling*, vol. 28, no. 4, pp. 2013–2021.

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1 SARIMAX Forecasts

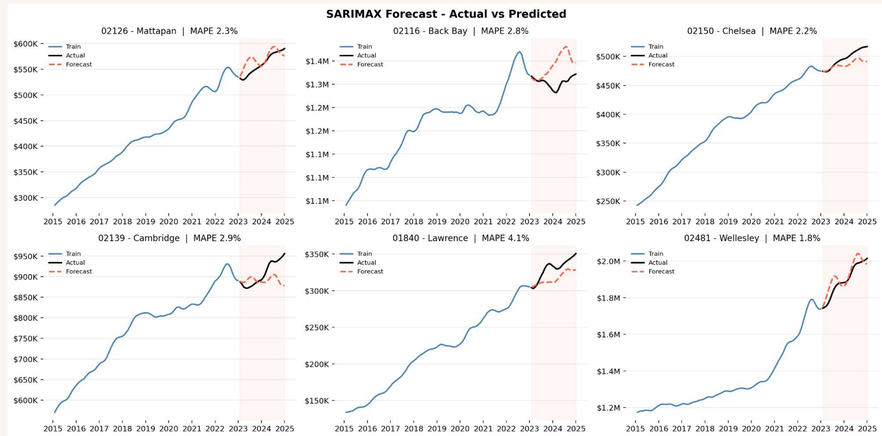

Fig. 1: Actual vs. forecast for six ZIP codes spanning Boston City, inner suburbs, and outer suburbs.

ZIP Code	MAPE	MAE
02481 Wellesley	1.8%	$32,510
02150 Chelsea	2.2%	$10,989
02126 Mattapan	2.3%	$12,612

ZIP Code	MAPE	MAE
02116 Back Bay	2.8%	$35,860
02139 Cambridge	2.9%	$25,041
01840 Lawrence	4.1%	$12,267

2 MLP Classifier: Stress Across Neighborhoods

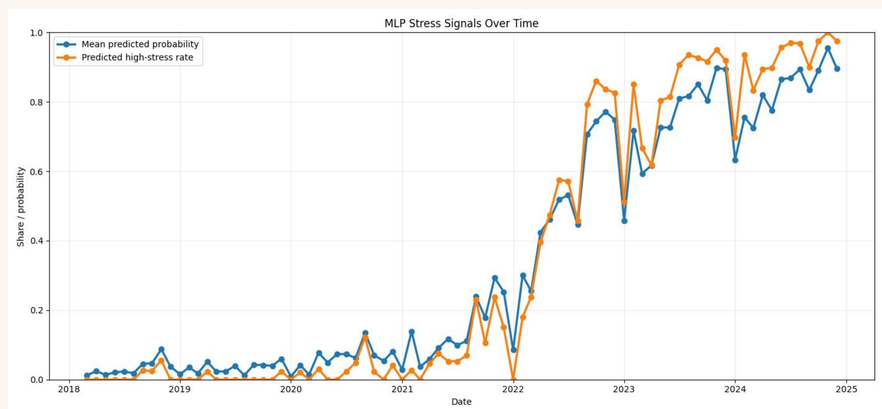

Fig. 2: Mean predicted stress probability (blue) and share of ZIP codes classified high stress (orange).

Both signals hover near zero through 2020, rise in 2021, and climb sharply through 2022 – 2023. By late 2024 the model flagged ∼**90% of ZIP codes** as high stress, including historically accessible neighborhoods like **Mattapan, Dorchester, and Roxbury Crossing** that lower and middle income residents have traditionally relied on.

Confusion Matrix

	Pred. Low	Pred. High
True Low	2,092	378
True High	82	786

Classification Metrics

Accuracy	**86.2%**
Precision	**0.68**
Recall	**0.91**
F1	**0.77**

3 Bayesian Regression

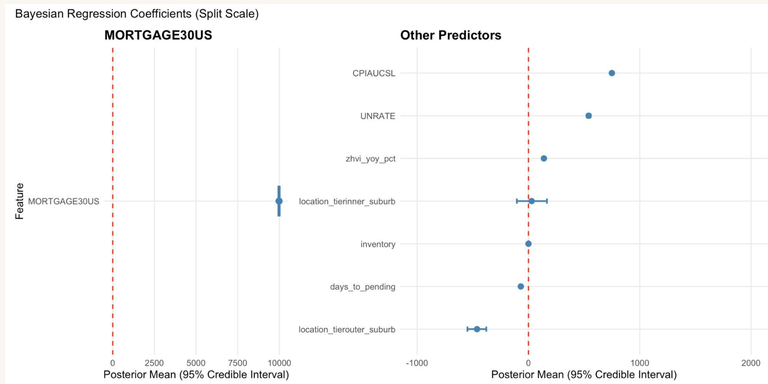

Fig. 3: Posterior means with 95% credible intervals. Mortgage rate dominates all others by an order of magnitude.

Test MAE	**$4,306**
Test RMSE	**$4,481**
Test MAPE	**2.35%**

Key Finding

Each +1 pp in the mortgage rate adds ∼$10,000 to the income needed to buy a home.

Conclusions & Future Work

The affordability crisis is no longer confined to expensive neighborhoods like Back Bay or Cambridge. It has spread into areas that lower and middle income residents have historically relied on. By late 2024 the MLP classifier flagged nearly **90% of ZIP codes** as high stress. The Bayesian model pinpoints the mechanism: each percentage point on the mortgage rate translates to roughly **$10,000 more income** required to buy a home, with high posterior certainty. The SARIMAX forecasts confirm the shift was large enough to challenge even well-specified time series models, particularly in high-value markets that experienced sharp post-2022 corrections.

A return to lower mortgage rates would be the single most impactful lever for easing stress, but without increased housing supply the underlying price pressure will persist.

Limitations: All three models predict *market-level* pressure but cannot identify which households within a ZIP code are at risk or have already been displaced. The income-needed variable is metro-level, flattening real variation across neighborhoods.

Future work: Incorporate ZIP-level demographic and income data to predict whether current residents can *afford to stay*, not just whether a neighborhood is expensive. Test Bayesian model robustness on a pre-2018 window to confirm that the mortgage rate dominance is not specific to the post-pandemic period.